

Advanced computing projects - Topics

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Storage system for optimizing distributed AI training in HPC

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Deep Learning (DL) training workloads (e.g., for image classification) often require large datasets comprising thousands to millions of samples. HPC centers are a natural fit to train these models given the available distributed and specialized compute power. However, when multiple DL jobs run on the same cluster, they compete for shared storage resources, namely the Parallel File System (PFS). Reading a large set of small files can easily saturate the PFS, leading to I/O performance degradation and even unavailability.

The goal of this project is to propose a solution that reduces AI training time while minimizing load on the PFS. The project should focus on distributed DL and leverage the resources of all nodes participating in the training workload. When applicable, techniques such as data replication and placement are also desirable to cache the same samples across multiple disks, further optimizing the training runtime.

References

- Accelerating Deep Learning Training Through Transparent Storage Tiering. Dantas, M., Leitão, D., Cui, P., Macedo, R., Liu, X., Xu, W., & Paulo, J. In 22nd IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGrid). IEEE, 2022.
- Clairvoyant prefetching for distributed machine learning I/O. Nikoli Dryden, Roman Böhringer, Tal Ben-Nun, and Torsten Hoefler. 2021. In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC '21). 2021

Storage system for optimizing LLM checkpointing in HPC

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Large Language Models (LLMs) training is a complex workload that can take several weeks to complete. The complexity and long execution time are inherent to their size, as they can easily reach billions of parameters that need to be refined across training. Upon a failure (e.g., one or more nodes used in the training process become unavailable), if these parameters are not saved, the training process needs to be restarted from scratch. To address this challenge, the model's state (i.e., activations, weights) is periodically checkpointed to persistent storage, such as the Parallel File System (PFS) in HPC environments. However, storing multiple checkpoints in a shared storage system can lead to performance degradation, especially when the storage system is being used by multiple jobs at the same time.

The goal of this work is to propose a solution that improves checkpointing performance of LLM training (without compromising correctness), while reducing the I/O pressure at the PFS. The project should consider different fault models (i.e., failure of multiple nodes) and leverage the resources of all nodes participating in the LLM training.

References

- Accelerating Deep Learning Training Through Transparent Storage Tiering. Dantas, M., Leitão, D., Cui, P., Macedo, R., Liu, X., Xu, W., & Paulo, J. In 22nd IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGrid). IEEE, 2022.
- Mohan, Jayashree, Amar Phanishayee, and Vijay Chidambaram. "{CheckFreq}: Frequent,{Fine-Grained}{DNN} Checkpointing." In 19th USENIX Conference on File and Storage Technologies (FAST 21), pp. 203-216. 2021.

Characterization of the energy profile of distributed HPC jobs

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HPC systems are increasingly constrained by energy consumption, which directly impacts operational costs, sustainability, and system scalability. While modern HPC infrastructures execute complex distributed jobs across heterogeneous resources (CPUs, GPUs, interconnects, and storage), understanding how energy is consumed by real applications remains challenging. Energy behavior can vary significantly depending on application characteristics, communication patterns, and runtime configurations, and these effects are often invisible to users and developers. As energy efficiency becomes a first-class concern in HPC system design and operation, there is a growing need for systematic, application-level studies that quantify and explain the energy footprint of distributed workloads.

The goal of this project is to perform a performance and energy characterization study of distributed HPC applications. The applications to be studied (*i.e.*, scientific workloads and AI training) will be selected and provided based on execution logs from the MACC and TACC operational teams, ensuring that the workloads are representative of real production environments, while the energy measurement tool will be provided by the supervision team.

References

- Anthony, L. F. W., Kanding, B., & Selvan, R. (2020). Carbontracker: Tracking and predicting the carbon footprint of training deep learning models. *arXiv preprint arXiv:2007.03051*.
- Hè, H., Friedman, M., & Rekatsinas, T. (2024). Energat: Fine-grained energy attribution for multi-tenancy. *ACM SIGENERGY Energy Informatics Review*, 4(3), 18-25
- <https://codecarbon.io/>

Performance study of key-value stores for HPC

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The HPC software stack offers multiple I/O libraries and abstractions, ranging from the traditional POSIX file system interface to higher-level parallel I/O frameworks designed to improve performance and portability. Despite these advances, many HPC applications still rely heavily on file-based I/O, which can become a scalability and performance bottleneck, especially for metadata-intensive or fine-grained access patterns. In contrast, key-value stores have been widely adopted in other domains to efficiently handle such workloads, offering alternative data access models that decouple data organization from file system semantics. However, the potential of key-value stores to improve I/O performance in HPC workloads has received limited attention, and their suitability for HPC execution patterns remains poorly understood.

The goal of this project is to conduct a comprehensive experimental study evaluating the performance suitability of key-value stores (e.g., WiredTiger, RocksDB) for HPC workloads, identifying opportunities and current limitations. The project will analyze representative I/O patterns commonly found in HPC applications and compare the behavior of key-value stores against traditional storage approaches (i.e., PFS, I/O libraries) under these workloads. The study will involve tuning key-value stores for HPC workloads, as well as designing experiments, selecting appropriate benchmarks, and systematically evaluating performance metrics such as throughput, latency, and resource utilization.

References

- <https://github.com/facebook/rocksdb/>
- Cao, Z., Dong, S., Vemuri, S., & Du, D. H. (2020). Characterizing, modeling, and benchmarking {RocksDB}{Key-Value} workloads at facebook. In *18th USENIX Conference on File and Storage Technologies (FAST 20)* (pp. 209-223)
- Kim, J., Lee, S., & Vetter, J. S. (2017, November). PapyrusKV: A high-performance parallel key-value store for distributed NVM architectures. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis* (pp. 1-14).
- Luo, C., & Carey, M. J. (2020). LSM-based storage techniques: a survey. *The VLDB Journal*, 29(1), 393-418

Carbon footprint of ARM architectures for High Energy Physics processing

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The dual constraints of computational throughput and operational sustainability increasingly define the evolution of High-Performance Computing (HPC). As the energy demands of exascale systems approach the multi-megawatt scale, the transition toward energy-efficient Instruction Set Architectures (ISAs) has become a critical research frontier. ARM-based processors, exemplified by the Fujitsu A64FX with its Scalable Vector Extensions (SVE) and the burgeoning NVIDIA Grace-Blackwell platform, offer a high-efficiency alternative to traditional CISC architectures. These technologies leverage advanced power gating, high-bandwidth memory (HBM), and superior performance-per-watt to mitigate the environmental impact of large-scale data centres. Reducing the carbon intensity of these infrastructures is imperative to align fundamental scientific research with global decarbonization targets and ensure that supercomputing infrastructures remain ecologically viable.

This work aims to conduct a systematic evaluation of diverse ARM-based hardware deployments across the European EuroHPC ecosystem to quantify their efficacy in processing High Energy Physics workloads. The project will benchmark critical data-intensive tasks—such as Monte Carlo event generation and large-scale signal reconstruction—across several prominent European testbeds (detailed in the table below). A primary objective is to analyse the relation between time-to-solution and energy-to-solution, specifically investigating how hardware-level features, such as Dynamic Voltage and Frequency Scaling and Power Capping, impact and can be leveraged to modulate consumption of these applications. By identifying the optimal configuration for "carbon-per-event" processing, this work will provide a framework for physicists to select appropriate hardware and tailor their computational strategies, balancing scientific precision with a minimised carbon footprint.

References

- Pereira, A., & Proença, A.J. (2021). HEP-Frame: Improving the efficiency of pipelined data transformation & filtering for scientific analyses. *Comput. Phys. Commun.*, 263, 107844.
- Simakov, N.A., Deleon, R.L., White, J.P., Jones, M.D., Furlani, T.R., Siegmann, E., & Harrison, R.J. (2023). Are we ready for broader adoption of ARM in the HPC community: Performance and Energy Efficiency Analysis of Benchmarks and Applications Executed on High-End ARM Systems. *Proceedings of the HPC Asia 2023 Workshops*.

Carbon-aware strategies for intra- and inter-application scheduling

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Mitigating the operational carbon footprint in High-Performance Computing (HPC) requires transitioning from traditional performance-centric scheduling to carbon-aware resource orchestration. While hardware efficiency provides the foundation for green computing, the temporal and spatial distribution of workloads—managed by system schedulers such as Slurm—offers a significant, often untapped, lever for decarbonization. By integrating marginal carbon intensity (CI) predictions into the scheduling logic, HPC centres can implement demand shifting (delaying non-critical tasks to periods of high renewable penetration) and demand shaping (modulating the execution profile based on energy grid signals). This approach acknowledges that a Watt consumed during a solar peak carries a fundamentally lower carbon penalty than one consumed during a fossil-fuel-intensive peak, thus decoupling scientific throughput from environmental degradation over a 24-hour cycle.

The objective of this research is to design and evaluate intra- and inter-application scheduling strategies that minimise total carbon emissions while adhering to strict execution time constraints. At the inter-application level, the project will develop and test a policy-driven framework for Slurm that prioritises job allocation based on real-time grid telemetry and hardware-specific power profiles. Simultaneously, intra-application strategies will leverage workload scheduling, fine-grained hardware power knobs, such as Dynamic Voltage and Frequency Scaling and software-defined power capping, to optimise the energy-to-solution of individual jobs. The success of these strategies will be measured using a composite objective function that balances carbon footprint and completion time. By simulating these strategies on European supercomputing testbeds, the work will provide a blueprint for next-generation, environmentally responsive middleware.

References

- Chen, J., He, Y., Zhang, Y., Han, P., & Du, C. (2022). Energy-aware scheduling for dependent tasks in heterogeneous multiprocessor systems. *J. Syst. Archit.*, 129, 102598.
- Fan, K., D'Antonio, M., Carpentieri, L., Cosenza, B., Ficarelli, F., & Cesarini, D. (2023). SYnergy: Fine-Grained Energy-Efficient Heterogeneous Computing for Scalable Energy Saving. *SC23: International Conference for High Performance Computing, Networking, Storage and Analysis*, 1-14.

FUSE-based File System for Small File Workloads in HPC

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With the AI boom, many HPC workloads are now generating and accessing a large quantity of small files (e.g., in DL training, when using Python virtual environments, etc). This new metadata-intensive pattern has been generating severe performance contention at shared storage systems across HPC users (e.g., Parallel File Systems, Network File Systems). This contention results in a bad experience to users, who see their jobs getting slower, and ultimately it can lead to unavailability of the whole cluster.

The goal of this project is to develop a userspace-based filesystem, using the FUSE framework, to alleviate such performance pressure under small-file workloads. Briefly, the main idea is to use FUSE to create a large-file abstraction that can contain several small files. This way, users can continue to deploy their jobs without any changes, but accesses to small files done by user workloads are translated into operations to large files persisted at the shared HPC storage systems. The solution is expected to reduce the number of metadata operations to the PFS, while ensuring good performance and maintaining crash-consistency.

References

- SafeFS: A Modular Architecture for Secure User-Space File Systems (One FUSE to rule them all)- Pontes, R., Burihabwa, D., Maia, F., Paulo, J., Schiavoni, V., Felber, P., Mercier, H., & Oliveira, R. In International Systems and Storage Conference (SYSTOR). 2017.
- To FUSE or Not to FUSE: Performance of User-Space File Systems. Bharath Kumar Reddy Vangoor, Vasily Tarasov, Erez Zadok. USENIX Conference on File and Storage Technologies (FAST '17). 2017.
- LibFuse repository - <https://github.com/libfuse/libfuse>

Adaptive DataLoader Configuration for AI Training in HPC

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AI training workloads executed in HPC environments rely on data-loading mechanisms to feed training samples into the compute pipeline. Modern AI frameworks expose several configuration parameters for data loaders, such as the number of worker processes, prefetch depth, and batch size. These parameters are typically defined statically before execution, even though their optimal values depend on workload characteristics and runtime system conditions.

The goal of this project is to investigate adaptive configuration strategies for AI data loaders in HPC environments. The proposed approach will monitor runtime metrics such as iteration time, data loading latency, and resource utilization, and dynamically adjust data loader parameters during training. By adapting these parameters at runtime, the project aims to improve training throughput and reduce performance variability without modifying the underlying training algorithm.

References

- Lewis, Noah, Jean Luca Bez, and Suren Byna. "I/o in machine learning applications on hpc systems: A 360-degree survey." *ACM Computing Surveys* 57, no. 10 (2025): 1-41.
- Ofeidis, Iason, Diego Kiedanski, and Leandros Tassiulas. "An overview of the data-loader landscape: Comparative performance analysis." In *2024 IEEE International Conference on Big Data (BigData)*, pp. 360-367. IEEE, 2024.
- Zhao, Juntao, Qi Lu, Wei Jia, Borui Wan, Lei Zuo, Junda Feng, Jianyu Jiang et al. "OVERLORD: Ultimate Scaling of DataLoader for Multi-Source Large Foundation Model Training." *arXiv preprint arXiv:2504.09844* (2025).

Adaptive Checkpoint Frequency Control for AI Training in HPC

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AI training workloads running in HPC environments rely on periodic checkpointing to preserve progress in case of failures. These checkpoints are commonly written at fixed intervals to shared storage systems. When multiple jobs checkpoint simultaneously, this behavior can generate significant I/O pressure and negatively impact performance for other workloads sharing the same system.

The goal of this project is to investigate adaptive checkpointing strategies for AI training workloads in HPC systems. Instead of using a fixed checkpoint interval, the project will explore simple policies that adjust checkpoint frequency based on observed execution behavior, such as checkpoint duration or training progress.

The project will be evaluated using small-scale AI training workloads and synthetic failure scenarios to assess the trade-offs between checkpoint overhead and recomputation cost.

References

- Strati, Foteini, Michael Friedman, and Ana Klimovic. "PCcheck: Persistent Concurrent Checkpointing for ML." In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 1, pp. 811-827. 2025.
- Yao, Chenxuan, Feifan Liu, Yuchong Hu, Zhengyu Liu, Xinjue Zheng, and Wenxiang Zhou. "LowDiff: Efficient Frequent Checkpointing via Low-Cost Differential for High-Performance Distributed Training Systems." In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis, pp. 1113-1126. 2025.
- Huang, Zimeng, Hao Nie, Haonan Jia, Bo Jiang, Junchen Guo, Jianyuan Lu, Rong Wen, Biao Lyu, Shunmin Zhu, and Xinbing Wang. "FlowCheck: Decoupling Checkpointing and Training of Large-Scale Models." In Proceedings of the Twentieth European Conference on Computer Systems, pp. 1334-1349. 2025.

Reinforcement Learning for Adaptive Cache Configuration

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Modern systems rely on multi-layer storage hierarchies that include caches and buffering mechanisms to reduce data access latency. These cache layers often expose configuration parameters such as cache size allocation, eviction policy, and prefetching behavior. In practice, these parameters are typically configured statically, even though optimal settings depend on workload access patterns and may change over time.

The goal of this project is to investigate reinforcement learning–based strategies for adaptive cache configuration. The proposed approach will monitor runtime metrics such as cache hit rate, access latency, and throughput, and dynamically adjust cache parameters during execution. By framing cache configuration as a sequential decision-making problem, the project aims to improve overall data access performance and reduce performance variability compared to static configurations.

References

- Zhou, Yang, Fang Wang, Zhan Shi, and Dan Feng. "An end-to-end automatic cache replacement policy using deep reinforcement learning." In Proceedings of the International Conference on Automated Planning and Scheduling, vol. 32, pp. 537-545. 2022.
- Zhou, Yang, Fang Wang, Zhan Shi, and Dan Feng. "An efficient deep reinforcement learning-based automatic cache replacement policy in cloud block storage systems." IEEE Transactions on Computers 73, no. 1 (2023): 164-177.
- CacheLib: Facebook's open source caching engine.
<https://engineering.fb.com/2021/09/02/open-source/cachelib/>

Exploring disaggregated memory for efficient computing

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Memory disaggregation is gaining traction with technologies like CXL that enable large, shared memory pools accessible across nodes. However, this approach introduces new trade-offs related to latency, bandwidth, and placement policies that affect application performance. This project will investigate these trade-offs by using and extending an existing memory-pool simulator capable of modeling multiple virtual memory pools with configurable latency, bandwidth, and topology. The students will enhance the simulator to support queueing effects, link contention, and NUMA-like distance classes, and will design policy modules for tiering and data placement, adaptive migration, bandwidth-aware scheduling, and admission control. The evaluation will involve both synthetic microbenchmarks (e.g., STREAM, LMBench) and application kernels (e.g., graph analytics, key-value stores) to measure how different policies and configurations influence performance under varying workloads.

The expected outcomes include the testing of an extensible open-source simulator for heterogeneous memory pools, a quantitative design space exploration identifying when memory disaggregation improves capacity utilization versus when it incurs performance penalties, and practical recommendations for policy design and runtime mechanisms. The work should contextualize its results against existing work on disaggregated memory and RDMA-based paging, showing how simulator-level decisions map to OS and runtime-level implementations such as NUMA placement or swapping strategies. This research aims to provide both a tool and insights to guide future system architects and researchers in leveraging memory disaggregation effectively.

References

- Wang, Z., Guo, Y., Lu, K., Wan, J., Wang, D., Yao, T., & Wu, H. (2024). Rcmp: Reconstructing RDMA-Based Memory Disaggregation via CXL. *ACM Transactions on Architecture and Code Optimization*, 21, 1 - 26.
- Harrington, E.D. (2025). Dynamic Memory Disaggregation for High-Performance Computing Using RDMA-Aware Schedulers. *Darpan International Research Analysis*.